autocompletion and inline documentation hints in Integrated Development Environments (IDEs). It enables safer and faster refactoring with type-aware tools, thereby reducing the risk of introducing errors during code changes. Additionally, it allows for the early detection of potential bugs, such as incorrect prop types or invalid state updates, which ensures higher code quality. Figure 3.5 demonstrates these benefits with a practical example, where TypeScript catches a type error when attempting to assign a number to `setIsRunning`, which expects a boolean value.

Figure 3.5 TypeScript type error when assigning an incorrect value type to setIsRunning.